MyDesign Portfolio

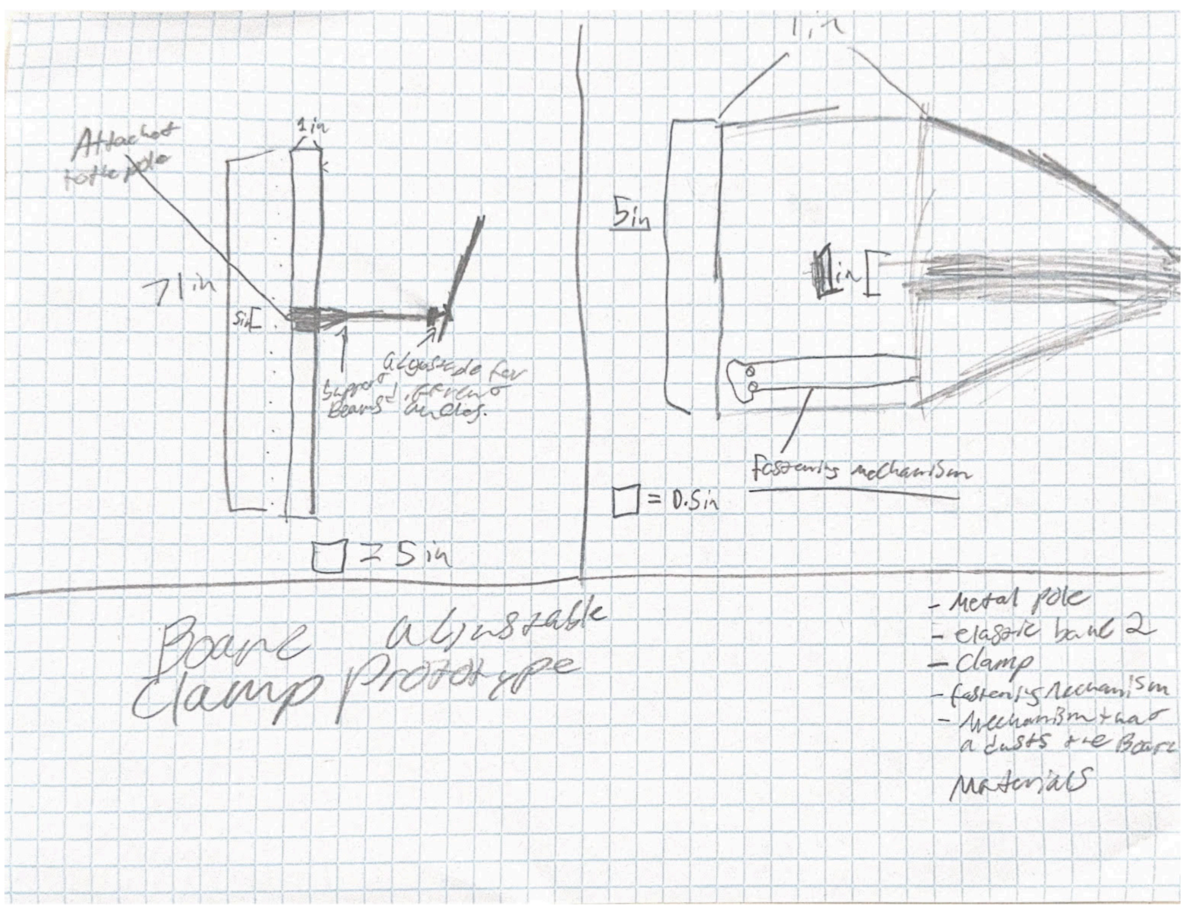
Guidance before filling out on the next page:

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

Design Idea #1--

Date: 3/10/25
Description: An adjustable clamp utilizing a pole like structure in the middle to extend away from the board, additionally the angle of the board holder is variable with the base being able to change direction with ease.



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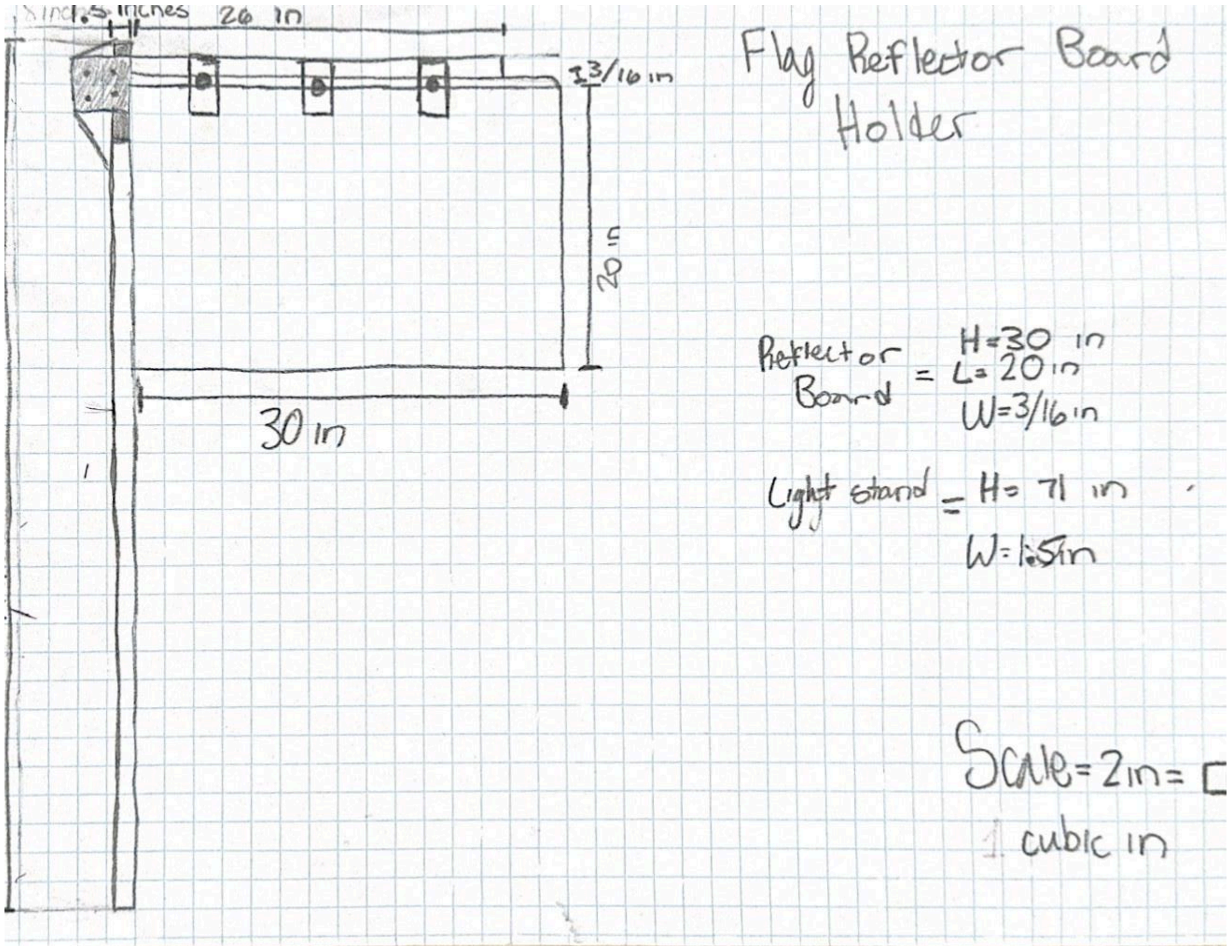
Design Idea #2 - Flag Reflector Board Holder

Date: 3/10/2025

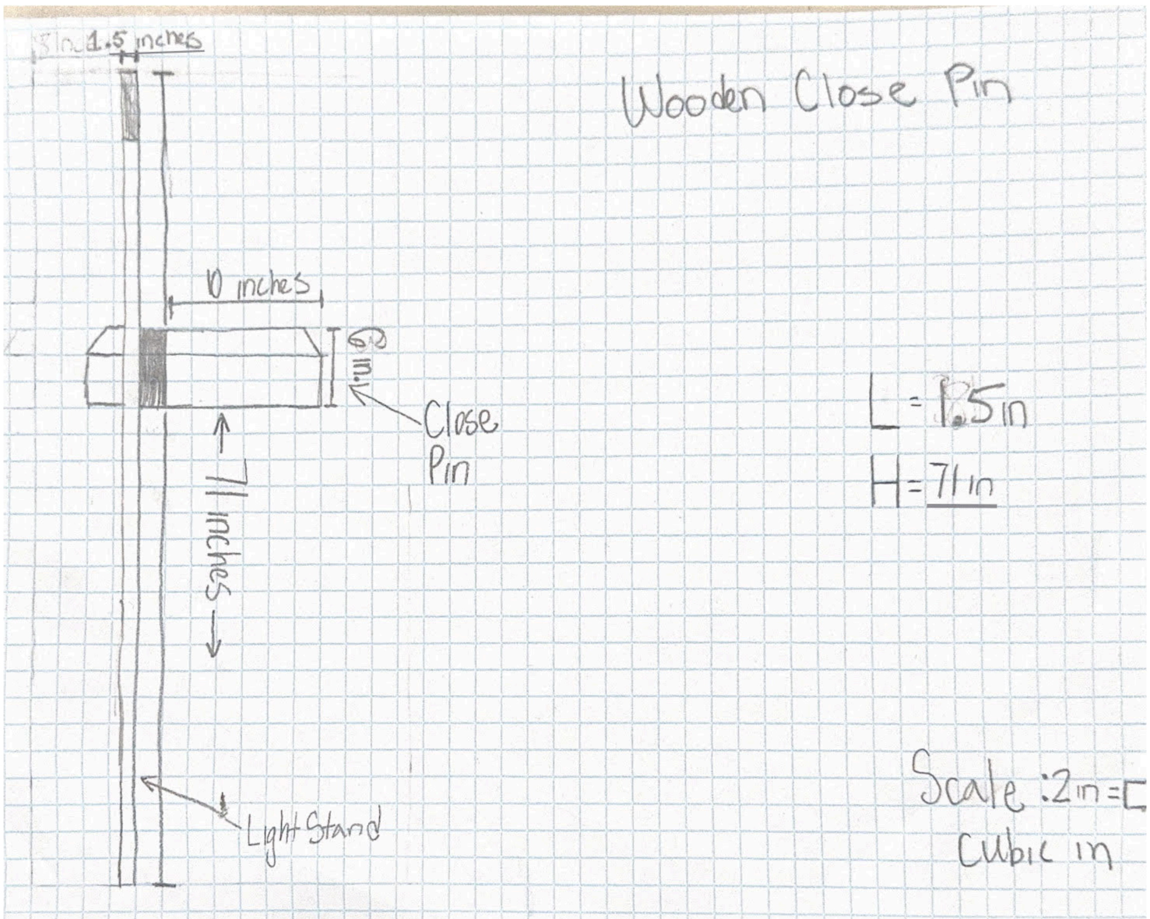
Description: A pole like attachment that attaches to the main pole of the tripod stand. The pole utilizes clips on the side of it to easily open and close to hold the reflecting board firmly in place. This model also allows for easy clip on/off access.



Design Idea #3 - Wooden Close Pin

Date: 3/10/25

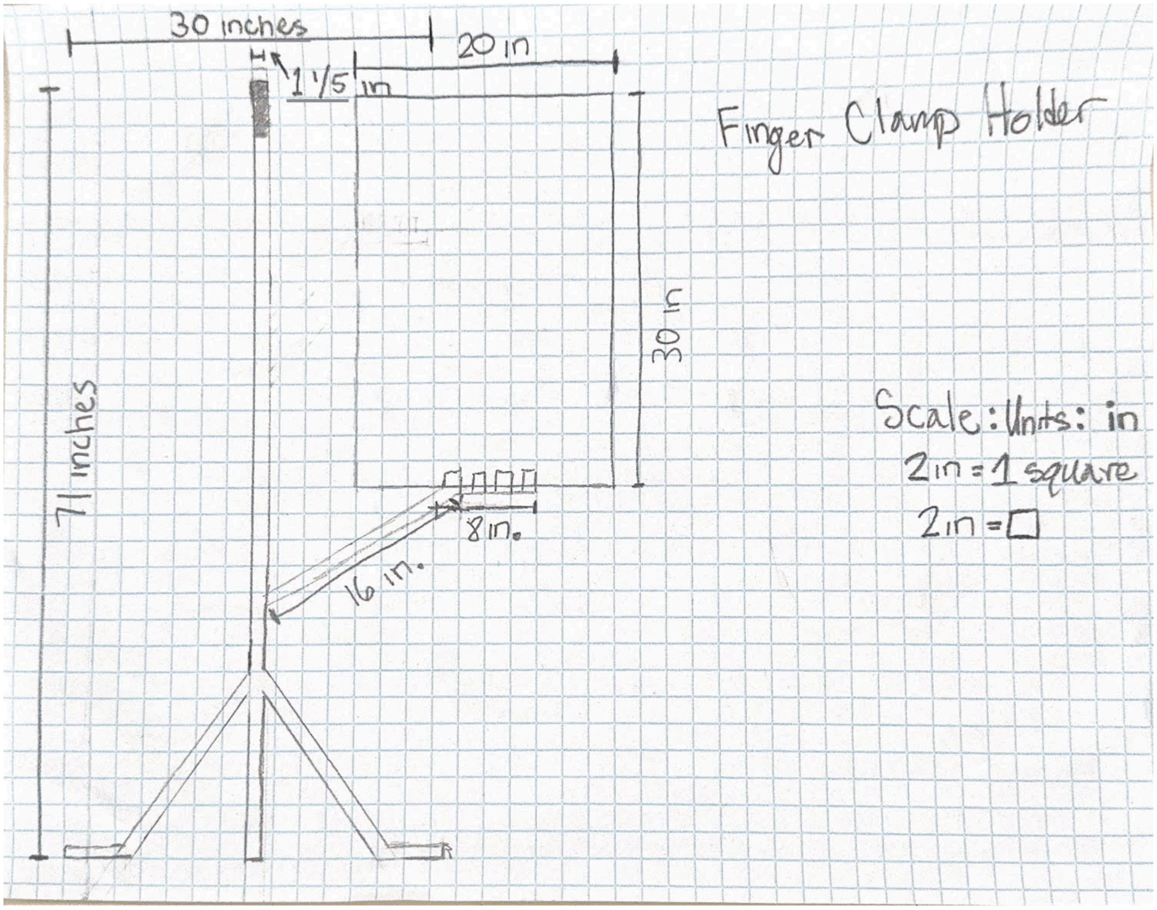
Description: This design takes inspiration from the household closepin, the mechanism utilizes the close pins design being able to close and clamp tightly, with easy on and off access.



Design Idea #4 - Finger Clamp Holder

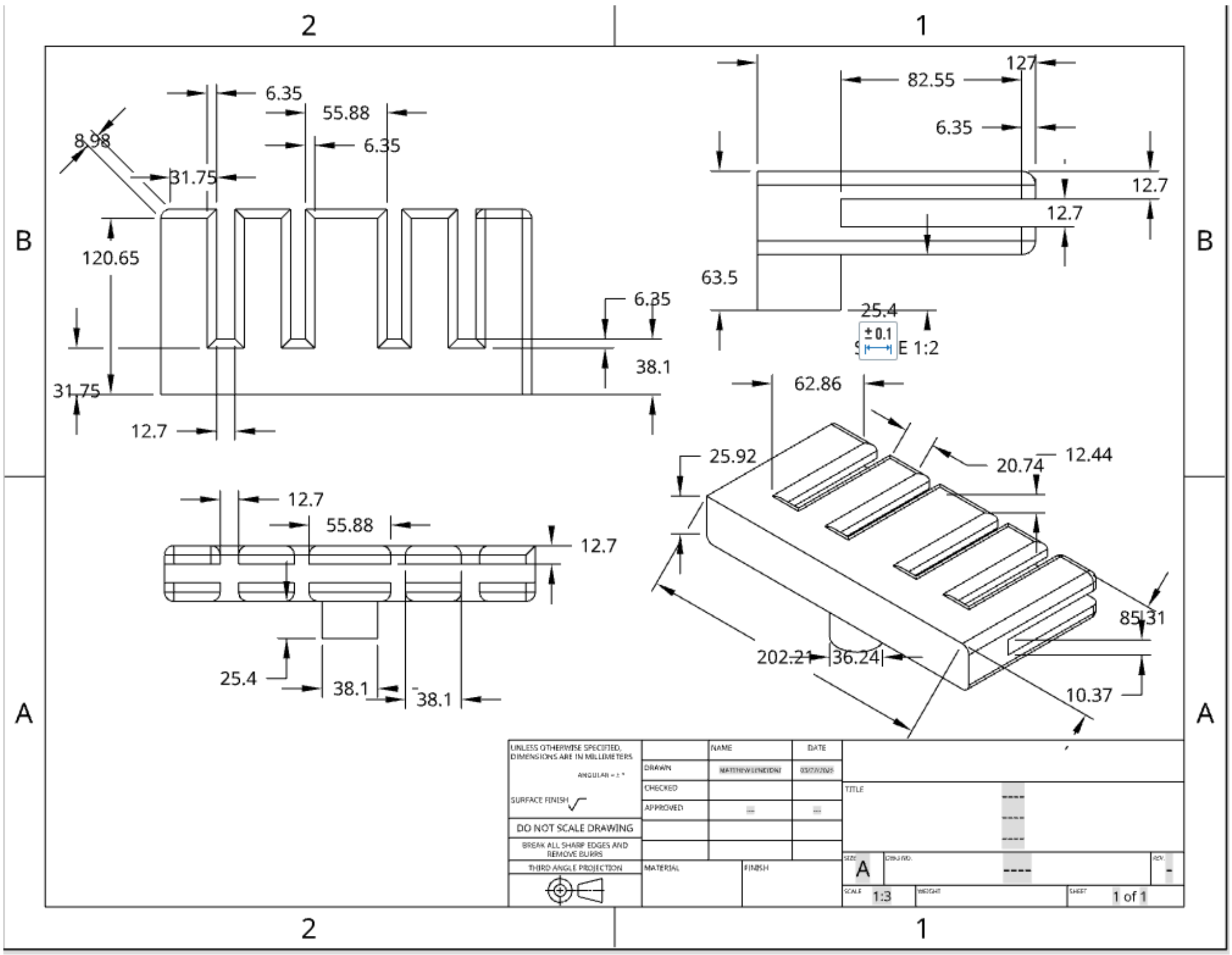
Date: 3/11/25

Description: This attachment is a very viable option as it has a pole like structure to extend away from the board and ensure sturdiness. Additionally the board is held with a finger shaped mechanism, allowing for the board to be snug, meanwhile it will easily be able to be taken on and off, and with such a small holder is a very cost effective model.



Design Idea #5 - Double Finger Clamp - OnShape

Date: 3/11/25
Description: This design is similar to the last one using a pole like structure to extend away from the board and ensuring sturdiness. Additionally the board is held with a finger shaped mechanism, allowing for the board to be snug meanwhile it will easily be able to be taken on and off. The main difference is the size of the bed for the board. With the larger bed it ensures an immense amount of sturdiness.



Weighted Ratings:

	Weight	Design 1	Design 2	Design 3	Design 4	Design 5
Weight	15%	0.6	0.15	0.3	0.45	0.75
Sturdiness	30%	0.9	0.6	1.2	0.3	1.5
Cost	10%	0.4	0.3	0.2	0.1	0.5
Flexibility	25%	0.5	0.75	1	0.25	1.25
Durability	20%	0.6	0.4	0.2	1	0.8
Scores:	100%	3	2.2	2.9	2.1	4.8
Ranking		4	2	3	1	5

Ranking Scale of 5-1, 5 is the best and 1 is the worst.

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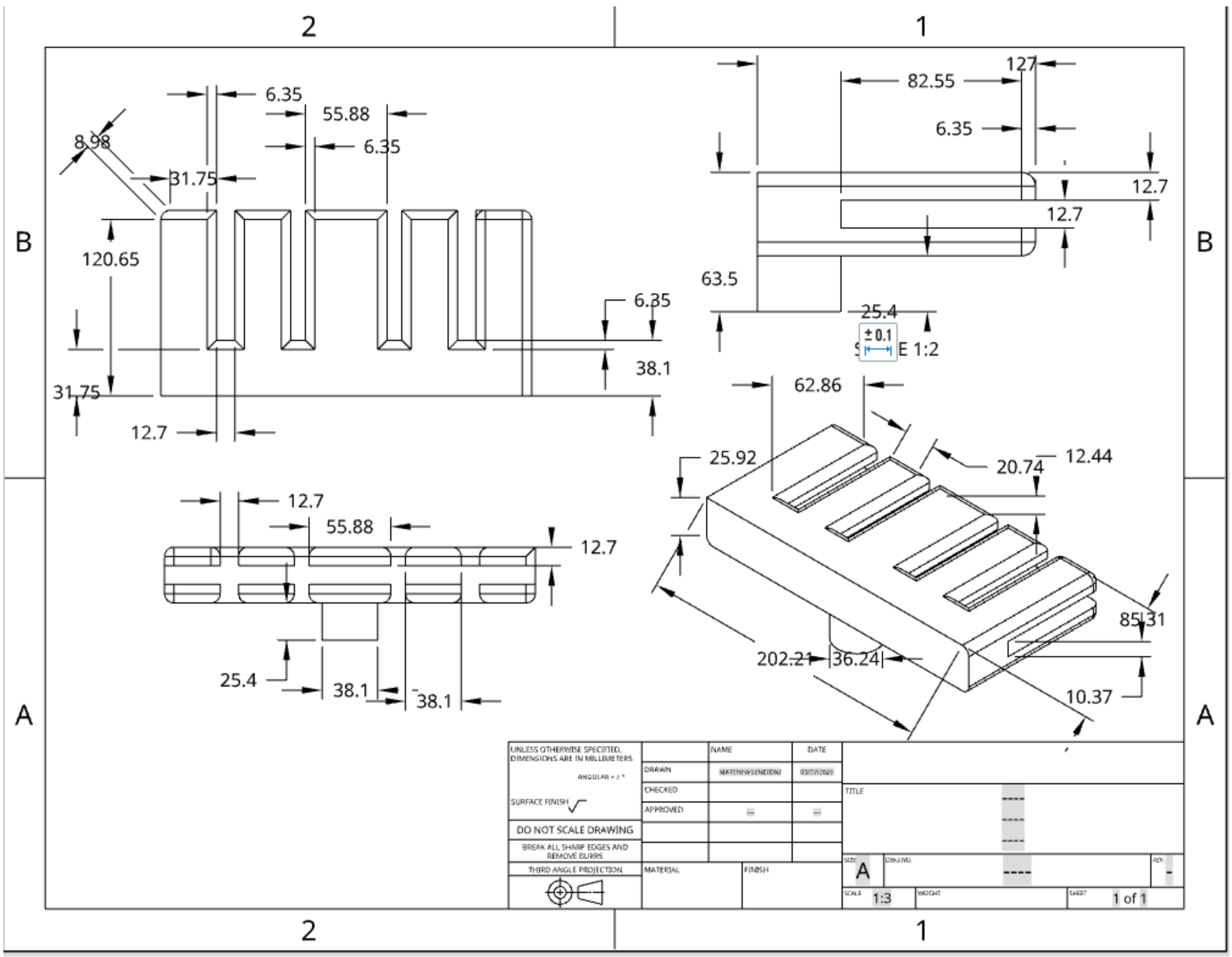
Design Blueprint #1

Date: 3/11/25

This mechanism will allow for a board to be slid into the finger like molds. The small “stump” will be held onto tightly by a clamp, (attached below) that will also clamp to a pole. Not seen in the picture is rubber bands which will add resistance and minimize movement of the board.



Manufacturing Methods:

- Step 1: 3d Printing
- Step 2: Hand Made

Step 3: Industrial Machinery
Supply List and Costs:

Item	**Cost each**	**# needed**	**Total**
PLA Filament	$20.00	1 (technically zero because the school has some available for use)	$20.00 ($0.00)
Neewer Double Super Clamp Camera Mount with Magic Arm	$19.99 (on sale)	1	$19.99
Hot Glue	$3.00	1 (pack of sticks)	$3.00
Duct Tape	$3.94	1 (roll)	$3.94
Total Cost			$21.94

Plan of Action *(D3)*

[Amazon.com : Neewer Double Super Clamp Camera Mount with Magic Arm, Cold Shoes for Desk Studio Light Stand Holder Photography Reflector, Flag, Cross Bars, Motorcycle, Umbrella, Pole Stick Shoot Accessories Tools]